

Lab 11 Report

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Due: 04/28/2026 @ 11:00 AM

1 Complexity

Using the Gauss-Jordan elimination method, the solver requires $\mathcal{O}(n^3)$ operations for n unknowns. This means that the overall complexity for this project is $\mathcal{O}(n^3)$. Parsing the input file is $\mathcal{O}(n)$, and constructing the matrix is $\mathcal{O}(n^2)$, so they are negligible when compared to the solver.

2 Loop Analysis

The loops with the most runtime happen in the Gauss-Jordan elimination process, which uses three nested loops over the number of unknowns. The matrix construction uses nested loops over nodes and edges, giving a complexity of $\mathcal{O}(n^2)$. One loop invariant is that, during elimination, columns become zero except at the diagonal. Another loop invariant is that, during back substitution, solved variables stay unchanged. Non-dominant loops include the parsing and file output loops.

3 Sources

AERE361_LabManual_11.pdf

<https://www.geeksforgeeks.org/compiler-design/introduction-of-parsing-ambiguity-and-parsers-set-1/>

<https://www.geeksforgeeks.org/linux-unix/bash-scripting-introduction-to-bash-and-bash-scripting/>